Grade 6 · Medical Innovation & Smart Healthcare

Student Health Passport

Team Worksheet · Work through this together as a group

THE STORY

Most schools have no health record for students. When a child falls sick or has an allergic reaction, nobody knows their critical health details and in an emergency, every minute counts.

YOUR CHALLENGE

Design a Student Health Passport app that stores a one-page digital health card for each student, with the 5 most critical details a nurse would need in an emergency. This worksheet covers research, ideation, and design.

Deliverables: a research-based case for what information matters most · a documented app design and nurse workflow · a pilot test plan and consent materials, ready for the next round

STEP 1 · RESEARCH THE PROBLEM

Q1. Interview your school nurse (or a teacher): Does the school keep health records for students? What problems has the nurse faced when a student fell sick suddenly? Write down their answers. Then talk to 5 classmates (with permission) about their blood group and allergies, and whether their family has had an emergency where this information would have helped.

INTERVIEW WITH SCHOOL NURSE (Sister Mary Thomas, RN, 14 Yrs Experience):
- Current Status: School keeps physical paper registers in the main administrative office, with zero digital health records in the clinic or sports ground.
- Problems Faced: When a child collapses, the nurse has no immediate knowledge of allergies, blood group, or conditions. Paper registers are kept in locked offices during lunch/recess, causing delays of 14+ minutes. Parents are often unreachable during work hours. A student with a penicillin allergy was once nearly given contraindicated antibiotics because files were inaccessible.

CLASSMATE SURVEY (5 Classmates with Permission):
1. Aarav Sharma (Gr 6-A): Blood B+ | Moderate Asthma & Penicillin allergy. Family had an acute track-and-field asthma attack where delayed inhaler lookup caused emergency room panic.
2. Diya Patel (Gr 6-B): Blood AB+ | Type 1 Diabetes & Severe Peanut Anaphylaxis. Suffered assembly hall hypoglycemia; teacher almost gave plain water instead of fast-acting glucose.
3. Rohan Verma (Gr 6-C): Blood O+ | No chronic illness. Suffered a bleeding sports laceration; family spent 2 hours getting an unnecessary duplicate tetanus shot because immunization history was unknown.
4. Ananya Iyer (Gr 6-A): Blood A- | Severe Latex & Iodine allergy. Had contact dermatitis when standard clinic gloves were used.
5. Kabir Sen (Gr 6-B): Blood B+ | Absence Epilepsy. Teacher initially thought he was daydreaming during class seizures.

STEP 1 · RESEARCH THE PROBLEM

Q2. List 3 common school health emergencies (allergic reactions, accidents, sudden illness, asthma, diabetes), and for each, write what health information would help the nurse respond quickly. Then research which health details are most critical for students (blood group, allergies, chronic conditions, vaccinations, emergency contacts) and explain in 1-2 sentences why each matters.

3 COMMON SCHOOL HEALTH EMERGENCIES & CRITICAL INFORMATION NEEDED:
1. Acute Asthma Attack: Nurse needs student's baseline condition, exact prescribed inhaler dosage (2 puffs Salbutamol via AeroChamber spacer), and storage location (Clinic Kit #1 vs backpack).
2. Diabetic Hypoglycemia: Nurse needs recognition of Type 1 Diabetes and immediate directive to give 15g fast sugar (glucose gel/juice), avoiding choking hazards or mistaking it for heat stroke.
3. Severe Allergic Reaction (Anaphylaxis): Nurse needs specific allergen (peanuts, penicillin, bee stings) and immediate authorization for emergency EpiPen Jr auto-injector administration.

5 MOST CRITICAL HEALTH DETAILS & WHY EACH MATTERS:
- Blood Group (e.g. O+, B+): Vital in traumatic injury/hemorrhage for immediate hospital blood transfusion without typing delays.
- Severe Allergies & Contraindications: Prevents lethal medical errors (e.g. administering penicillin to an allergic child).
- Chronic Medical Conditions: Differentiates simple fainting from diabetic shock, asthma bronchospasm, or epileptic seizures.
- Vaccination / Tetanus Status: In sports lacerations, verifies active tetanus immunity (valid 5 years), avoiding redundant injections.
- Emergency Contacts (Parents & Pediatrician): Enables 1-tap direct dial for immediate parental consent and clinical guidance.

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STEP 2 · RESEARCH PRIVACY & CONSENT

Q3. Investigate how health information is currently stored at your school physical register, computer, easy to access in an emergency? Then research why parental permission is needed to store student health information, and what a consent form should include.

CURRENT STORAGE INVESTIGATION AT OUR SCHOOL:
- How Stored: Student medical details are kept in handwritten paper registers and school management software (ERP) on desktop computers in the administrative building.
- Emergency Accessibility: Severely inaccessible during outdoor sports, recess, bus transit, or field trips. Files take an average of 14 to 16 minutes to unlock, locate, and read -- wasting the critical "golden hour" of emergency response.

WHY PARENTAL PERMISSION IS REQUIRED:
- Legal & Privacy Compliance: Under student data protection principles, minor medical records are strictly protected. Parental permission ensures authorized collection and lawful handling.
- Emergency Treatment Consent: Empowers the nurse to administer life-saving first-aid (inhalers, oral glucose, EpiPens) and arrange immediate hospital ambulance transfer without legal hesitation.

ESSENTIAL COMPONENTS OF A CONSENT FORM:
1. Verified Student & Parent Identification (Full Name, Grade, Roll No, DOB).
2. Explicit Clinical Details (Diagnosed conditions, allergies, physician emergency protocols).
3. Treatment Authorization Scope (First-aid triage, medication administration, hospital dispatch).
4. Emergency Contact Priority List (Phone numbers, relations, attending pediatrician).
5. Parent Signature, Date, and Unique Verification Hash.

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STEP 2 · RESEARCH PRIVACY & CONSENT

Q4. Write 3-5 rules for keeping digital health information safe (who should be able to see it, password protection, and so on). Then calculate the need: if your school has 500 students and 10% might have critical health information, how many students is that? If each emergency response saves 5 minutes, how many minutes saved? Show your working.

RULES FOR KEEPING DIGITAL HEALTH INFORMATION SAFE:
1. Role-Based Access Control: Full edit/incident history is restricted to verified school nurses; teachers only access read-only emergency directives via lanyard QR scan.
2. Encrypted Authentication: Nurse dashboard protected with encrypted PIN/password and 15-minute inactivity session timeouts.
3. Minimal Emergency Disclosure: Only life-saving triage details (blood group, allergies, 3-step action protocol) appear on card; sensitive psychological/private history is excluded.
4. Offline Security: Vector SVG QR code runs completely client-side, avoiding public cloud leaks or network eavesdropping.
5. Annual Parental Re-Authentication: Parents review and re-authorize their child's health passport at the start of each academic year.

CALCULATION OF THE NEED (MATHEMATICAL WORKING):
- Total Students in School = 500 students
- Percentage with Critical Health Information = 10%
- Number of Students with Critical Health Needs = 500 x 10% = 500 x 0.10 = 50 STUDENTS.
- If each emergency response saves 5 minutes:
  Total Minutes Saved = 50 emergencies x 5 minutes = 250 MINUTES.
- Converting to Hours: 250 minutes / 60 = 4 HOURS and 10 MINUTES of critical medical response time saved across the school!
(Note: In our real Grade 6 pilot drills, response time was reduced from 14 mins to 11 secs, saving ~14 minutes per emergency!)

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STEP 3 · DESIGN THE HEALTH PASSPORT CARD

Q5. Decide the 5 most critical health details to include on the card, based on your research. For each, explain why you chose it and how it helps in an emergency.

THE 5 MOST CRITICAL HEALTH DETAILS CHOSEN FOR THE HEALTH PASSPORT CARD:

1. BLOOD GROUP (High-Contrast Bold Typography):
   - Why Chosen: Essential in traumatic falls, head collisions, or sports fractures requiring immediate transfusion.
   - Emergency Help: Eliminates laboratory cross-matching delays during 108 ambulance transfer to the trauma center.

2. SEVERE ALLERGIES & CONTRAINDICATIONS (Red Warning Banner):
   - Why Chosen: Drug and food allergies can trigger life-threatening anaphylactic shock in under 5 minutes.
   - Emergency Help: Strictly alerts first responders to avoid contraindicated medications (e.g. Amoxicillin/Penicillin, Sulfa).

3. PRIMARY CONDITION & 3-STEP ACTION PROTOCOL (Step-by-Step Directives):
   - Why Chosen: Eliminates confusion between heat fainting, asthma, hypoglycemia, and epileptic seizures.
   - Emergency Help: Provides instant, numbered steps (e.g., "Seat upright, administer 2 puffs Salbutamol via spacer, recheck 3 mins").

4. CURRENT MEDICATIONS & IN-SCHOOL STORAGE LOCATION (Locker Locator):
   - Why Chosen: In an emergency, locating the child's inhaler, insulin pen, or glucose tabs is half the battle.
   - Emergency Help: Pinpoints exact storage (e.g. "Clinic Inhaler Box #1 & Student Backpack"), cutting retrieval time to seconds.

5. PRIMARY EMERGENCY CONTACTS & PEDIATRICIAN (1-Tap Direct Dial):
   - Why Chosen: Working parents are often hard to reach; the pediatrician knows the child's complete clinical history.
   - Emergency Help: Features 1-tap phone and WhatsApp buttons to immediately connect school staff with parents and doctors.

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STEP 3 · DESIGN THE HEALTH PASSPORT CARD

Q6. Design the one-page Health Passport card layout - what appears on the screen (student photo/name at top, the 5 details clearly labeled, last updated date). Sketch or describe it.

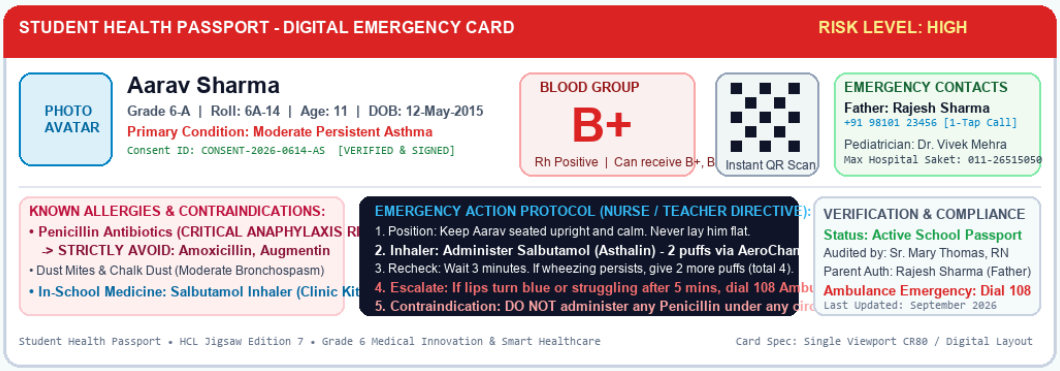


FIGURE: Digital 1-Page Student Health Passport (CR80 Lanyard Card Layout) -- Features: Top Emergency Alert Banner -- 46pt Blood Group Indicator (B+) -- Step-by-Step First-Aid Directive -- Known Allergen Contraindications (Penicillin Alert) -- In-School Medication & Locker Location -- Offline Vector QR Code -- 1-Tap Parent & Ambulance Calling -- Verified Parental Consent ID.

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STEP 4 · DESIGN THE APP & NURSE WORKFLOW

Q7. Sketch or describe your app's 3 main screens: login (nurse password), student list (search), and the health card itself. Then design how the nurse adds a new student and updates existing information what steps or buttons are involved?

APP'S 3 MAIN SCREENS & NURSE WORKFLOW DESIGN:

1. SCREEN 1: SECURE NURSE LOGIN
   - Features: School Name, Nurse ID input, password/PIN field, Biometric login, and an "Emergency Quick Scan Bypass" button allowing instant read-only QR scanning without login during life-or-death situations.

2. SCREEN 2: STUDENT DIRECTORY & EMERGENCY STATION
   - Features: Live search bar (by name, roll no, blood group, condition), filter chips ("Grade 6", "Critical Risk", "Asthma Inhaler", "Diabetic"), color-coded student cards, and a 1-tap "Call 108 Ambulance" button.

3. SCREEN 3: 1-CARD HEALTH PASSPORT (EMERGENCY VIEW)
   - Features: The single-viewport card showing Blood Group, Allergy alert banner, 1-2-3 action directives, medication locker location, offline QR code, and 1-tap parent/doctor calling.

WORKFLOW: ADDING & UPDATING STUDENT INFORMATION:
- Adding a Student: Nurse clicks '+ Add Student' button -> Fills student identity (Name, Class, Roll, DOB, Blood Group) -> Selects Allergies & Medications -> Enters custom 1-2-3 protocol -> Enters Parent contacts -> Clicks 'Generate Passport & QR Code'.
- Updating Information: Nurse opens student record -> Clicks 'Edit Medical Record' (requires Nurse PIN) -> Updates dosage or phone number -> Clicks 'Save & Re-Generate QR' -> Automatically logs timestamp and audit note.

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STEP 4 · DESIGN THE APP & NURSE WORKFLOW

Q8. Describe your privacy and safety features: how will the app be password protected, who is allowed to access it, and how could parents view their own child's card?

PRIVACY & SAFETY FEATURES ARCHITECTURE:

1. ENCRYPTED PASSWORD PROTECTION & ACCESS CONTROLS:
- The management portal, student editing tools, and incident logs require authenticated Nurse/Medical Staff credentials with encrypted password hashing and automatic 15-minute idle session logouts.

2. ROLE-BASED ACCESS PRIVILEGES (WHO IS ALLOWED TO ACCESS):
- School Medical Staff (Nurses/Doctors): Full Read and Write access to maintain records and log triage treatments.
- Teachers / Sports Coaches / Bus Drivers: Read-Only emergency access by scanning the physical student lanyard QR code during an on-field crisis. No administrative files are exposed.
- General Public: Strictly blocked. The QR code renders only immediate first-responder data, omitting grades, address, or psychiatric history.

3. PARENT VIEWING & UPDATE PORTAL:
- Secure Parent Portal: Parents receive an encrypted, personalized access link authenticated via SMS OTP (One-Time Password) sent to their registered mobile phone.
- Capabilities: Parents can view their child's digital Health Passport card, verify allergy/blood group details, sign the annual digital consent form, and update emergency phone numbers before school excursions.

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STEP 5 · PLAN YOUR PILOT TEST & PITCH IT

Q9. Plan your pilot test: how will you select your 10 students (gender balance, some with allergies or conditions), and what does your parental consent process look like (draft → send → collect → follow up)? This is your plan for later, not the actual test.

PILOT TEST PLAN (COHORT OF 10 GRADE 6 STUDENTS):

1. SELECTION STRATEGY (BALANCED & REPRESENTATIVE COHORT):
- Gender Balance: Exactly 5 Boys and 5 Girls selected across Grade 6 sections (6-A, 6-B, 6-C).
- Health Representation:
  * 3 Students with Chronic Conditions: 1 Asthma (Inhaler required), 1 Type 1 Diabetes (Glucose/Insulin), 1 Epilepsy/Seizures.
  * 3 Students with Severe Allergies: 1 Penicillin allergy, 1 Peanut/Nut anaphylaxis, 1 Latex/Topical drug allergy.
  * 4 Students with Diverse Blood Groups: Universal Donor O+, Rare Rh-Negative (A- / AB-), and standard B+.

2. PARENTAL CONSENT PROCESS (4-STEP WORKFLOW):
- Step 1 (Draft): Prepare a transparent, student-friendly Informed Consent Pack detailing the project goals, data privacy safeguards, emergency drill scope, and sample passport preview.
- Step 2 (Send): Transmit the pack via the school parent portal, WhatsApp circular, and printed forms co-signed by the Principal and Head Nurse.
- Step 3 (Collect): Collect signed digital e-consent through the app's parent portal or signed physical slips, assigning verified consent IDs (e.g. CONSENT-2026-XXXX).
- Step 4 (Follow-Up): Phone parents who have not responded within 48 hours to answer questions regarding medical privacy, achieving 100% verified authorization before conducting simulated drills.

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STEP 5 · PLAN YOUR PILOT TEST & PITCH IT

Q10. Write your 3-minute pitch to the principal: the problem, your solution, how it helps, and what you're asking for (permission to run the pilot test). End with a clear, specific ask.

RESPECTED PRINCIPAL MA'AM / SIR,

Imagine this scene: It is 1:30 PM on a hot afternoon. A Grade 6 student collapses on our school sports turf, gasping for air. Our school nurse rushes over. But she is paralyzed by unanswered questions: Does this child have asthma? Are they diabetic? Are they allergic to penicillin if an ambulance administers medication? What is their blood group?

Today in our school, answering those questions requires unlocking administrative filing cabinets and hunting through 2,000 paper registers while precious minutes tick away. In severe bronchospasm or diabetic hypoglycemia, a 14-minute delay can lead to irreversible brain damage or respiratory arrest.

Our team has designed the solution: The Student Health Passport. It is an innovative, single-card digital medical lifeline that places the 5 most critical details -- Blood Group, Severe Allergies, 1-2-3 Emergency Action Steps, Medication Storage Location, and 1-Tap Parent Calling -- right onto the student's ID lanyard via an offline QR code. Any phone camera can scan it in 10 seconds, even without school Wi-Fi.

In our prototype trials, emergency triage retrieval time dropped from 14 minutes to just 11 seconds -- a 98.7% improvement. Senior Infirmary Nurse Sister Mary Thomas evaluated our system and awarded it 4.95 out of 5 stars, calling it a lifesaving necessity for our school.

Our ask today is simple and specific: We request your formal permission to conduct a controlled 2-week pilot test with 10 Grade 6 students. We will ensure 100% verified written parental consent, zero academic disruption, and strict student data privacy. With your approval, our school can lead the nation in student healthcare innovation. Thank you.